

MATH I (Fall 2026)

Week 1 NOTES

Part 1

Part 1 covers:

- The scope of the Math I course
- What is Mathematics, Mathematical Analysis, Calculus
- Why Math is important for Engineers
- How contemporary engineering jobs rank Math (among the knowledge domains)
- The important contributions of Uzbek scholars to science and math, and the dissemination of the decimal positional system
- The historical evolution of numeral (numeration) systems.

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Introduction: Mathematics, Mathematical Analysis, Calculus

The twenty-first century is witnessing one of the most profound technological transformations in human history. Artificial intelligence, scientific computing, digital twins, autonomous systems, advanced manufacturing, intelligent materials, and quantum technologies are reshaping engineering practice at an unprecedented pace. Engineers are no longer expected simply to design machines, circuits, or structures. They are increasingly asked to design intelligent systems that combine physical principles with math reasoning, computation, data, optimization, and learning.

Despite these remarkable advances, one discipline remains at the heart of modern engineering: **mathematical analysis**. Every engineering application, every optimization algorithm, every machine learning model, every control system, every finite element analysis, every AI model ultimately relies on the mathematical language developed by Newton, Leibniz, Cauchy, Weierstrass, and many others. In the age of artificial intelligence, mathematical analysis has become more essential than ever.

More on Mathematics, Mathematical Analysis and Calculus:

Mathematics is the broadest discipline, encompassing many fields such as algebra, geometry, probability, statistics, discrete mathematics, optimization, and mathematical analysis.

Mathematical Analysis forms one of its fundamental branches, providing the rigorous theory of continuous functions, limits, derivatives, integrals, and infinite processes. It provides the mathematical foundation for understanding how systems evolve over space and time. From the motion of planets to the training of artificial intelligence models, Mathematical Analysis offers the language and tools needed to describe, predict, and optimize continuous phenomena.

Calculus is the most application-oriented part of Mathematical Analysis. Differential calculus studies rates of change, slopes, optimization and approximation, while integral calculus studies accumulation, areas, volumes and total effects. Calculus is therefore closely related to mathematical analysis but normally emphasizes calculations, interpretation and applications more than formal proofs.

Note: In this course, we may use the terms “mathematical analysis” and “calculus” indistinguishably.

In the Math I course, our goal is not only to teach students how to compute derivatives and integrals, but also to help them understand why these ideas transformed science and engineering. Engineering has always been the discipline that transforms scientific knowledge into reliable, safe, efficient, and useful technologies. In the AI era, the role of mathematics is expanding. Mathematics is no longer merely the language of science; it has become one of the fundamental languages of intelligence itself. Mathematical Analysis, as the mathematics of continuous transformation, lies at the center of this new landscape.

This course is intended for first-year engineering students, and its underlying philosophy is that *students should learn mathematics not only as a collection of methods and techniques, but as a way of thinking about the world*. This course is about understanding how continuous systems are represented, approximated, analyzed, optimized, and ultimately used to solve real engineering problems and shape the modern world. We aim to help students see mathematical analysis not as an obstacle to overcome, but as one of humanity's greatest intellectual achievements and one of the most powerful tools available to every engineer in the age of modern/advanced engineering and artificial intelligence.

A few more things about Calculus...

Unlike mathematics as a whole, which studies arbitrary abstract structures, **calculus focuses on continuous structures**. Before calculus, mathematics was largely about static objects: numbers, triangles, equations, algebraic expressions.

Calculus introduced something fundamentally different: **Processes**. Instead of asking "What is this object?", calculus asks "**How is this object changing?**" and "**What accumulates as it changes?**" That's an enormous conceptual leap. As **nature changes continuously** (at least to a useful approximation), and real-world engineering systems involve continuous motion, continuous temperature, continuous voltage, continuous stress, etc., **calculus provides the math to analyze the continuous world**. [\[see, at the end of this file, a very recent result on the limitations of this statement\]](#)

Calculus addresses two fundamental questions...

1. How fast is something changing?

This leads to the *derivative*. Examples: velocity, acceleration, heat flow, electric current, learning rate, population growth. The derivative measures **local change**.

2. How much has accumulated?

This leads to the *integral*. Examples: distance traveled, energy consumed, mass, electric charge, total cost. The integral measures **global accumulation**.

... and unifies change and accumulation. We are going to discuss this later in Math I, when we will present the Fundamental Theorem of Calculus, which says that *the rate of accumulation equals the accumulated rate*.

Calculus studies infinitesimal change and connects local-to-global reasoning

The key innovation was thinking locally. Instead of analyzing an entire curve, calculus asks "What happens at this instant?" The derivative is essentially an infinitely local measurement. The integral reconstructs the global picture from infinitely many local contributions.

Examples of how calculus connects local behavior with global behavior.

Local slope → Entire function

Local stress → Bridge stability

Local heat flow → Temperature distribution

Local learning rule → Neural network intelligence

Why Math is important for the Engineer - what do the jobs say?

Recent research work (Neofotistos, 2026 [working paper]; Asteris & Neofotistos, *Leadership and Management in Engineering*, 2012, 12(1)) identifies math as one of the most central knowledge in engineering jobs, by analyzing the weighted jobs-knowledge network, for occupations in the Engineering job family of the O*NET system, USA's main occupational dataset. The original analysis, based on O*NET 31.0 and the O*NET-SOC taxonomy, found Mathematics to be among the four most central knowledge domains in the areas of Materials & Nanotechnology, Electrical & Computing Engineering, Power and Energy Engineering, and Industrial Engineering & Management (see Table below). Strengthening education in the identified central knowledge domains empowers the integration of knowledge across engineering jobs.

*Table: Ranking top knowledge domains in main engineering fields (based on data from O*NET)*

Materials & Nanotechnology	Electrical & Computer	Power & Energy	Industrial Eng. & Management
Engineering and Technology	Engineering and Technology	Engineering and Technology	Engineering and Technology
Mathematics	Computers and Electronics	Design	Design
Physics	Design	Mathematics	Mechanical
Design	Mathematics	Physics	Mathematics
Computers and Electronics	Physics	Computers and Electronics	Computers and Electronics
Chemistry	Mechanical	English Language	Production and Processing
English Language	English Language	Mechanical	Education and Training
Mechanical	Production and Processing	Administration and Management	Physics
Production and Processing	Education and Training	Chemistry	English Language
Education and Training	Administration and Management	Education and Training	Administration and Management

Historical background:

Uzbekistan's scientific heritage: A Historic Crossroads of Mathematics, Science, and Engineering

Uzbekistan's scientific heritage is not simply a matter of national pride, but an integral part of the global history of mathematics, science, and engineering.

Uzbekistan occupies a unique place in the history of science. Situated at the crossroads of the Silk Road, it became a meeting point for ideas from Greece, Persia, India, China, and the Islamic world, giving rise to one of history's greatest scientific traditions. Between the ninth and fifteenth centuries, scholars from cities such as Khwarazm, Bukhara, Samarkand, and Fergana made fundamental contributions to mathematics, astronomy, medicine, geography, engineering, and the natural sciences. Their work formed an essential bridge between the scientific achievements of antiquity and the development of modern science in Europe and Asia.

Among the most influential figures is **Muhammad ibn Musa al-Khwarizmi** (c. 780–850), whose name gave rise to the modern word **algorithm**, while his celebrated book *Al-Jabr wa'l-Muqabala* gave the world the term **algebra**. Al-Khwarizmi transformed mathematics from a collection of arithmetic techniques into a systematic discipline for solving equations and reasoning symbolically. His methods became the foundation of algebra, numerical computation, and, ultimately, modern computer science. Every computer algorithm, every software program, and every artificial intelligence system owes an intellectual debt to the mathematical tradition initiated by Al-Khwarizmi. ([Outsource](#))

The scientific tradition continued with **Ahmad al-Farghani (Alfraganus)** (c. 800–870), born in the Fergana Valley. His astronomical writings became standard references throughout the medieval world and were translated into Latin, influencing European astronomy for centuries. His work illustrates the close relationship between mathematics, astronomy, engineering, and precise measurement that continues to characterize modern scientific practice. ([birunijournal.uz](#))

Another towering figure was **Abu Rayhan al-Biruni** (973–1048), one of history's greatest polymaths. Born in Khwarazm, he made pioneering contributions to mathematics, astronomy, geodesy, geography, mineralogy, physics, and the scientific study of different cultures. Remarkably, Al-Biruni combined mathematical theory with careful observation and experimentation, measuring the Earth's radius with impressive accuracy and developing methods that anticipated aspects of the modern scientific method. His work demonstrates that mathematics is most powerful when united with careful observation of the natural world. ([uzbekistan.org](#))

In the fifteenth century, **Mirzo Ulugh Beg** (1394–1449) transformed Samarkand into one of the world's leading scientific centers. Rather than being remembered only as a ruler, Ulugh Beg should also be recognized as one of history's great astronomers and mathematicians. The observatory he established in Samarkand produced one of the most accurate star catalogues before the invention of the telescope and demonstrated the importance of combining mathematical theory with precise measurement and large-scale scientific collaboration. **Ghiyath al-Din Jamshid al-Kashi** (1380–1429), a distinguished mathematician, astronomer and scientific-instrument designer became a leading member of the

fifteenth-century scientific school established by Ulugh Beg in Samarkand. His influential work *The Key to Arithmetic* presented systematic methods for calculation and helped advance the practical use of decimal fractions, while his *Treatise on the Circumference* calculated π to an accuracy of approximately sixteen decimal places—an exceptional achievement for his period. Al-Kashi also made important contributions to trigonometry, including highly accurate values of $\sin 1^\circ$ and a clear formulation of the law of cosines, contributing significantly to Samarkand’s emergence as one of the world’s foremost centres of mathematics and astronomy.

Today, this remarkable scientific heritage provides an inspiring foundation for educating the next generation of engineers and scientists. The legacy of Al-Khwarizmi, Al-Biruni, Al-Farghani, Ulugh Beg, Ibn Sina, and many other scholars reminds us that mathematics is far more than a collection of formulas—it is a universal language for understanding nature, solving practical problems, and advancing human civilization. As Uzbekistan continues to invest in science, engineering, digital technologies, and artificial intelligence, it builds upon a tradition that has already shaped the intellectual history of the world. In many respects, the journey from **algebra** to **algorithms**, and from the astronomical observatories of Samarkand to today's AI laboratories and scientific computing centers, represents a continuous story of human curiosity, mathematical reasoning, and technological innovation.

A particularly important contribution:

The decimal system and the importance of Uzbek scholars in disseminating it

The **decimal positional system** was invented in India, but it became a global scientific language largely through the work of scholars of the Islamic Golden Age. Among them, Muhammad ibn Musa al-Khwarizmi, played a decisive role by explaining, formalizing, and disseminating the Hindu numeral system (through Latin translations of his work, Europe learned not only the new numerals but also systematic methods of calculation—algorithms—that remain at the heart of modern mathematics, computing, and artificial intelligence).

The decimal positional numeral system evolved over many centuries and through several civilizations. It is more accurate to distinguish four major stages:

Stage	Approximate Date	Contribution
Counting with fingers	Prehistory	Probably explains the widespread adoption of base 10.
Hindu decimal positional system	c. 200 BCE – 500 CE	Development of the positional base-10 system and the concept of zero.
Islamic Golden Age	8th–12th centuries	Systematic dissemination and refinement through scholars such as Al-Khwarizmi.
Europe	12th–16th centuries	Gradual replacement of Roman numerals by Hindu–Arabic numerals.

Let’s examine each stage:

1. Why base 10?

No one knows with certainty why humans adopted base 10. The most widely accepted explanation is physiological: Humans have 10 fingers. Many early civilizations counted using their fingers, making ten a natural base. Interestingly, not all cultures adopted base 10 (e.g., ancient Babylonians used base 60).

2. Early decimal systems (before positional notation)

The Egyptians (around 3000 BCE) already used a decimal counting system. However, it was not positional. For example, 100 was represented by a completely different symbol than 10. Repeating symbols indicated larger quantities. Similarly, Greek and Roman numerals were largely decimal in spirit but not positional. In Notes 2, we will discuss why Roman numbers are not good for doing math operations with large numbers.

3. The invention of the decimal positional system

The true breakthrough occurred in India, between roughly 200 BCE and 500 CE. Indian mathematicians gradually developed positional notation, place value, zero as a numeral, decimal arithmetic.

This is essentially the numeral system we still use today. This positional principle is one of the greatest inventions in the history of mathematics. Most historians therefore credit Indian mathematicians—rather than a single individual—with inventing the decimal positional numeral system.

4. Zero

Perhaps the greatest innovation was 0. Earlier civilizations understood "nothing" but did not always use zero as a numeral. Indian mathematicians introduced zero as a placeholder, a number, an arithmetic object. Without zero, modern arithmetic and computers would be impossible.

5. Al-Khwarizmi and dissemination

This is where Uzbekistan enters the story. Around 820 CE, **Muhammad ibn Musa al-Khwarizmi** wrote *On the Calculation with Hindu Numerals* (original Arabic title now lost; known through Latin translations). This book explained the Hindu decimal numeral system, decimal arithmetic, algorithms for calculation. Although the Arabic original no longer survives, its Latin translations introduced Europe to the new numeral system. Al-Khwarizmi's enormous contribution was explaining the decimal system, formalizing it, teaching it, disseminating it.

6. Why were the numerals called "Arabic numerals"?

Ironically, they were developed in India, but Europeans learned them mainly through Arabic mathematical texts. Thus, they became known in Europe as "Hindu-Arabic numerals" or simply "Arabic

numerals." Today historians generally use *Hindu-Arabic Numeral System* to acknowledge both contributions.

7. Fibonacci and Europe

The next major milestone came in 1202. Leonardo of Pisa (Fibonacci) published *Liber Abaci*. He learned the Hindu-Arabic numeral system while working around the Mediterranean. His book demonstrated its superiority over Roman numerals.

8. Slow adoption

Surprisingly, Europe did not immediately adopt the decimal system. Many merchants distrusted it. Roman numerals remained common for centuries.

Approximate timeline:

12th century (introduced) →

13th–15th centuries (used mainly by scholars and merchants) →

16th century (widely accepted) →

17th century (almost universal in scientific work.)

9. Why was it revolutionary?

The decimal positional system made possible efficient arithmetic, algebra, bookkeeping, commerce, astronomy, engineering, and science. Thus, the decimal system is arguably one of humanity's greatest technological inventions.

Concluding...: All the above, which constitute *Math I – Notes 1*, provide an introduction to math, math analysis, a historical evolution of *numeral systems*, highlighting also the importance of math for the modern engineer (as manifested in the high ranking of the math knowledge in the contemporary engineering jobs).

In *Math I - Notes 2*, we will present -in more detail- numeral (numeration) systems, such as the decimal system (which is the system that we commonly use, and we will use throughout this course), explaining why positional systems represent numbers in a way that is more efficient for doing math. We will also present the scientific and engineering notations used in representing very large or very small numbers.

In *Math I - Notes 3*, we will present number sets and number systems from natural numbers to integers, rational, real, and complex numbers.

Having said all that, let's start our Math I journey this Fall semester.

Real numbers, continuity and the Navier–Stokes result

Calculus is built on the continuous real-number line: between any two distinct real numbers there are infinitely many others, and limits can represent quantities approaching a value arbitrarily closely. **In engineering, this mathematical continuum allows us to describe temperature, pressure and velocity as functions defined at every point in space and time.** For example, instead of tracking individual water molecules, fluid mechanics represents a fluid by continuous fields such as velocity $u(x, t)$ and pressure $p(x, t)$. Their evolution is described by the Navier–Stokes equations:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad \nabla \cdot u = 0.$$

In September 2026, OpenAI announced an AI-generated proof that, for a particular three-dimensional, incompressible Navier–Stokes system with a smooth applied force, an initially smooth flow can develop a *finite-time singularity*: its mathematical velocity becomes unbounded even though its total energy remains finite. The proposed solution involves an increasingly stretched and concentrated vortex. This means that smooth initial data do not necessarily guarantee a smooth solution for all future times. The result was accompanied by a formal proof in Lean, but, as with any newly announced major theorem, it must still undergo detailed independent examination by the mathematical community. [OpenAI's announcement and proof](#)

This does not mean that real water can attain infinite velocity. Instead, **it means that the continuous mathematical model may cease to provide a physically meaningful description under the constructed conditions.** A real fluid consists of molecules and contains additional physical effects that the idealized equations do not represent. **The lesson for engineers is therefore important: continuity and smoothness make calculus and differential equations extraordinarily powerful, but every continuous model has assumptions and a domain of validity.** When a model predicts a singularity, the engineer must examine whether finer-scale physics, a particle-based description, or another model is required.

TO THE READER:

The Notes for Math I Week 1 have been prepared by Prof. Georgios Neofotistos, with excerpts (selected text) from his book (in preparation) titled *Calculus for the 21st-century Engineer - Mathematics, Computation, and Artificial Intelligence (AI)*. This work is licensed under a Creative Commons “Attribution-NonCommercial-NoDerivatives 4.0 International” license.

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